

*finite
adhesion*

a bounded number of vertices makes H planar except for a subgraph of bounded linear width sewn on to the unique cuff of $S^2 - 1$.) A tree-decomposition $(T, (V_t)_{t \in T})$ of a graph G has *finite adhesion* if all its adhesion sets are finite and for every infinite path $t_1 t_2 \dots$ in T the value of $\liminf_{i \rightarrow \infty} |V_{t_i} \cap V_{t_{i+1}}|$ is finite.

Unlike its counterpart for K^n , the excluded- $K^{\aleph_0}$ structure theorem has a direct converse. It thus characterizes the graphs without a $K^{\aleph_0}$ minor, as follows:

Theorem 12.6.8. *A graph G has no $K^{\aleph_0}$ minor if and only if G has a tree-decomposition of finite adhesion whose torsos are nearly planar.*

*finite
tree-width*

Finally, a structure theorem for excluding $K^{\aleph_0}$ as a topological minor. Let us say that G has *finite tree-width* if G admits a tree-decomposition $(T, (V_t)_{t \in T})$ into finite parts such that for every infinite path $t_1 t_2 \dots$ in T the set $\bigcup_{j \geq 1} \bigcap_{i \geq j} V_{t_i}$ is finite.

Theorem 12.6.9. *The following assertions are equivalent for connected graphs G :*

- (i) G does not contain $K^{\aleph_0}$ as a topological minor;
- (ii) G has finite tree-width;
- (iii) G has a normal spanning tree T such that for every ray R in T there are only finitely many vertices v such that G contains an infinite v – $(R - v)$ fan.

12.7 The graph minor theorem

Graph properties that are closed under taking minors occur frequently in graph theory. Among the most natural examples are the properties of being embeddable in some fixed surface, such as planarity.

By Kuratowski's theorem, planarity can be expressed by forbidding the minors K^5 and $K_{3,3}$. This is a *good characterization* of planarity in the following sense. Suppose we wish to persuade someone that a certain graph is planar: this is easy (at least intuitively) if we can produce a drawing of the graph. But how do we persuade someone that a graph is non-planar? By Kuratowski's theorem, there is also an easy way to do that: we just have to exhibit an IK^5 or $IK_{3,3}$ in our graph, as an easily checked 'certificate' for non-planarity. Our simple Proposition 12.6.2 is another example of a good characterization: if a graph has tree width < 3 , we can prove this by exhibiting a suitable tree-decomposition; if not, we can produce an IK^4 as evidence.

Theorems that characterize a property $\mathcal{P}$ by a set of forbidden minors are doubtless among the most attractive results in graph theory. As we saw in Lemma 12.6.1, such a characterization exists whenever $\mathcal{P}$ is minor-closed: then $\mathcal{P} = \text{Forb}_{\preceq}(\overline{\mathcal{P}})$, where $\overline{\mathcal{P}}$ is the complement of $\mathcal{P}$. However, one naturally seeks to make the set of forbidden minors as small as possible. And there is indeed a unique smallest such set: the set

$$\mathcal{K}_{\mathcal{P}} := \{ H \mid H \text{ is } \preceq\text{-minimal in } \overline{\mathcal{P}} \} \quad \text{Kuratowski set } \mathcal{K}_{\mathcal{P}} \quad (12.6.1)$$

satisfies $\mathcal{P} = \text{Forb}_{\preceq}(\mathcal{K}_{\mathcal{P}})$ and is contained in every other set $\mathcal{H}$ such that $\mathcal{P} = \text{Forb}_{\preceq}(\mathcal{H})$. We call $\mathcal{K}_{\mathcal{P}}$ the *Kuratowski set* for $\mathcal{P}$.

Clearly, the elements of $\mathcal{K}_{\mathcal{P}}$ are incomparable under the minor relation $\preceq$. Now the *graph minor theorem* of Robertson & Seymour says that any set of $\preceq$ -incomparable graphs must be finite:

Theorem 12.7.1. (Robertson & Seymour 1986–2004) *graph minor theorem*
The finite graphs are well-quasi-ordered by the minor relation $\preceq$.

We shall give a sketch of the proof of the graph minor theorem at the end of this section.

Corollary 12.7.2. *The Kuratowski set for any minor-closed graph property is finite.* □

As a special case of Corollary 12.7.2 we have, at least in principle, a Kuratowski-type theorem for every surface S : the property $\mathcal{P}(S)$ of embeddability in S is characterized by the finite set $\mathcal{K}_{\mathcal{P}(S)}$ of forbidden minors. $\mathcal{P}(S)$

Corollary 12.7.3. *For every surface S there exists a finite set of graphs $H_1, \dots, H_n$ such that a graph is embeddable in S if and only if it contains none of $H_1, \dots, H_n$ as a minor.* □

While Corollary 12.7.3 is immediate from the graph minor theorem, it can also be proved more directly. It is our next goal to do this. The main step is to prove that the graphs in $\mathcal{K}_{\mathcal{P}(S)}$ do not contain arbitrarily large grids as minors (Lemma 12.7.4). Then their tree-width is bounded (Theorem 12.6.3), so $\mathcal{K}_{\mathcal{P}(S)}$ is well-quasi-ordered (Theorem 12.4.2) and therefore finite.

The proof of Lemma 12.7.4 gives a good impression of the interplay between graph minors and surface topology, which – by way of Theorem 12.6.6, which we could not prove here – is also one of the key ingredients of the proof of the graph minor theorem. Appendix B summarizes the necessary background on surfaces, including a lemma

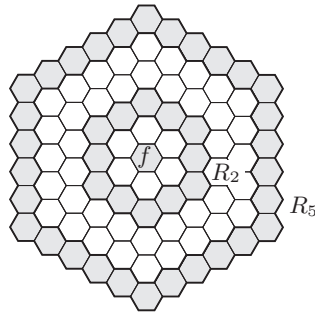


Fig. 12.7.1. The hexagonal grid H^6 with central face f and rings R_2 and R_5

used in the proof. For convenience (cf. Proposition 1.7.3 (ii)), we shall work with hexagonal rather than square grids.

H^r Denote by H^r the plane hexagonal grid whose dual has radius r (Figure 12.7.1). The face corresponding to the central vertex of its dual is its *central face*. (Generally, when we speak of the *faces* of H^r , we mean its hexagonal faces, not its outer face.) Call a subgrid H^k of H^r *standard* if their central faces coincide. We write S_k for the perimeter cycle of the standard subgrid H^k in H^r ; for example, S_1 is the hexagon bounding the central face of H^r . The *ring* R_k is the subgraph of H^r formed by S_k and S_{k+1} and the edges between them.

Lemma 12.7.4. *For every surface S there exists an integer r such that no graph that is minimal with the property of not being embeddable in S contains H^r as a topological minor.*

(4.1.2)
(4.2.2)
(4.3.2)
(App. B)

Proof. Let G be a graph that cannot be embedded in S and is minimal with this property. Our proof will run roughly as follows. Since G is minimally not embeddable in S , we can embed it in an only slightly larger surface S' . If G contains a very large H^r grid, then by Lemma B.6 some large H^m subgrid will be flat in S' , that is, the union of its faces in S' will be a disc D' . We then pick an edge e from the middle of this H^m grid and embed $G - e$ in S . Again by Lemma B.6, one of the rings of our H^m will be flat in S . In this ring we can embed the (planar) subgraph of G which our first embedding had placed in D' ; note that this subgraph contains the edge e . The rest of G can then be embedded in S outside this ring much as before, yielding an embedding of all of G in S (a contradiction).

More formally, let $\varepsilon := \varepsilon(S)$ denote the Euler genus of S . Let r be large enough that H^r contains $\varepsilon + 3$ disjoint copies of H^{m+1} , where $m := 3\varepsilon + 4$. We show that G has no TH^r subgraph.

Let $e' = u'v'$ be any edge of G , and choose an embedding σ' of $G - e'$ in S . Choose a face with u' on its boundary, and another with v'

on its boundary. Cut a disc out of each face and add a handle between the two holes, to obtain a surface S' of Euler genus $\varepsilon + 2$ (Lemma B.3). Embedding e' along this handle, extend σ' to an embedding of G in S' .

$$\begin{array}{rcl} \sigma': G & \hookrightarrow & S' \\ & & H \\ & & f \end{array}$$

Suppose G has a subgraph $H = TH^r$. Let $f: H^r \rightarrow H$ map the vertices of H^r to the corresponding branch vertices of H , and its edges to the corresponding paths in H between those vertices. Let us show that H^r has a subgrid H^m (not necessarily standard) whose hexagonal face boundaries correspond (by $\sigma' \circ f$) to circles in S' that bound disjoint open discs there.

By the choice of r , we can find $\varepsilon + 3$ disjoint copies of H^{m+1} in H^r . The standard subgrids H^m of these H^{m+1} are not only disjoint, but sufficiently spaced out in H^r that their deletion leaves a tree $T \subseteq H^r$ that sends an edge to each of them (Figure 12.7.2). Hence whenever we pick one hexagon from each of these H^m and delete the images C of those hexagons in S' , the component D_0 of the remainder of S' that contains $(\sigma' \circ f)(T)$ meets all those C in its boundary. By Lemma B.6 and $\varepsilon(S') = \varepsilon + 2$, therefore, it cannot be true that none of our circles C bounds a disc in S' that is disjoint from $(\sigma' \circ f)(T)$.

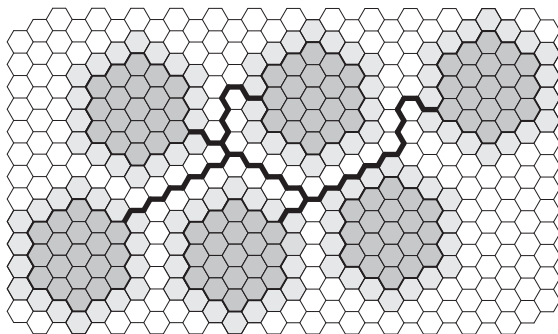


Fig. 12.7.2. Disjoint copies of H^m ($m = 3$) linked up by a tree in the rest of H^r

Hence for one of our copies of H^m in H^r , the image of every hexagon in S' bounds an open disc that is disjoint from $(\sigma' \circ f)(T)$. Let us show that these discs are disjoint. If not, then one of them, D say, contains a point x from the boundary of another such disc. But then D also contains $(\sigma' \circ f)(T)$, contrary to assumption, because we can walk from x to $(\sigma' \circ f)(T)$ in $(\sigma' \circ f)(H^r) \subseteq S'$ avoiding the boundary of D .

From now on, we shall work with this fixed H^m and will no longer consider its supergraph H^r . We write $C_i := f(S_i)$ for the images in G of the concentric cycles S_i of this H^m ($i = 1, \dots, m$).

 C_i

Pick an edge $e = uv$ of C_1 , and choose an embedding σ of $G - e$ in S . As before, Lemma B.6 implies that one of the $\varepsilon + 1$ disjoint rings R_{3i+2} in our H^m ($i = 0, \dots, \varepsilon$), R_k say, has the property that its hexagons

$$\begin{array}{rcl} \sigma: G - e & \hookrightarrow & S \\ & & e \\ & & k \end{array}$$

correspond (by $\sigma \circ f$) to circles in S that bound disjoint open discs there (Figure 12.7.3). Let $R \supseteq (\sigma \circ f)(R_k)$ be the closure in S of the union of those discs, which is a cylinder in S . One of its two boundary circles is the image under σ of the cycle $C := C_{k+1}$ in G to which f maps the perimeter cycle S_{k+1} of our special ring $R_k \subseteq H^m$.

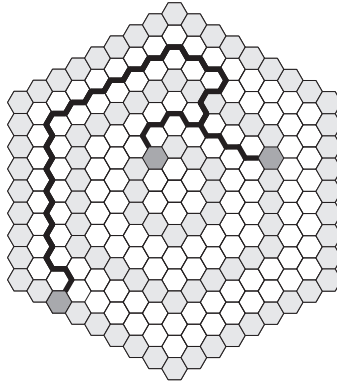


Fig. 12.7.3. A tree linking up hexagons selected from the rings $R_2, R_5, R_8 \dots$

Let $H' := f(H^{k+1}) \subseteq G$, where H^{k+1} is standard in our H^m . Recall that $\sigma' \circ f$ maps the hexagons of H^{k+1} to circles in S' bounding disjoint open discs there. The closure in S' of the union of these discs is a disc D' in S' , bounded by $\sigma'(C)$. Deleting a small open disc inside D' that does not meet $\sigma'(G)$, we obtain a cylinder $R' \subseteq S'$ that contains $\sigma'(H')$.

We shall now combine the embeddings $\sigma: G - e \hookrightarrow S$ and $\sigma': G \hookrightarrow S'$ to an embedding $\sigma'': G \hookrightarrow S$, which will contradict the choice of G . Let $\varphi: \sigma'(C) \rightarrow \sigma(C)$ be a homeomorphism between the images of C in S' and in S that commutes with these embeddings, i.e., is such that $\sigma|_C = (\varphi \circ \sigma')|_C$. Then extend this to a homeomorphism $\varphi: R' \rightarrow R$. The idea now is to define σ'' as $\varphi \circ \sigma'$ on the part of G which σ' maps to D' (which includes the edge e on which σ is undefined), and as σ on the rest of G (Fig. 12.7.4).

To make these two partial maps compatible, we start by defining σ'' on C as $\sigma|_C = (\varphi \circ \sigma')|_C$. Next, we define σ'' separately on the components of $G - C$. Since $\sigma'(C)$ bounds the disc D' in S' , we know that σ' maps each component J of $G - C$ either entirely to D' or entirely to $S' \setminus D'$. On all the components J such that $\sigma'(J) \subseteq D'$, and on all the edges they send to G , we define σ'' as $\varphi \circ \sigma'$. Thus, σ'' embeds these components in R . Since $e \in f(H^k) = H' - C$, this includes the component of $G - C$ that contains e .

It remains to define σ'' on the components of $G - C$ which σ' maps to $S' \setminus D'$. As $\sigma'(C_k) \subseteq D'$, these do not meet C_k . Since $\sigma(C \cup C_k)$ is

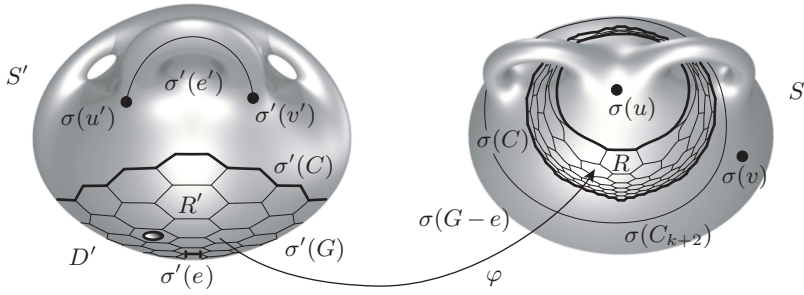


Fig. 12.7.4. Combining $\sigma': G \hookrightarrow S'$ and $\sigma: G - e \hookrightarrow S$ to $\sigma'': G \hookrightarrow S$

the frontier of R in S , this means that $\sigma(J) \subseteq S \setminus R$ or $\sigma(J) \subseteq R$ for every such component J .

For the component J_0 of $G - C$ that contains C_{k+2} we cannot have $\sigma(J_0) \subseteq R$: as $S_{k+2} \cap R_k = \emptyset$, this would mean that $\sigma(C_{k+2})$ lies in a disc $D \subseteq R$ corresponding to a face of R_k , which is impossible since S_{k+2} sends edges to vertices of S_{k+1} outside the boundary of that face. We thus have $\sigma(J_0) \subseteq S \setminus R$, and define σ'' as σ on J_0 and on all the J_0 - C edges of G .

Next, consider any remaining component J of $G - C$ that sends no edge to C . If $\sigma(J) \subseteq S \setminus R$, we define σ'' on J as σ . If $\sigma(J) \subseteq R$, then J is planar. Since J sends no edge to C , we can have σ'' map J to any open disc in R that has not so far been used by σ'' .

It remains to define σ'' on the components $J \neq J_0$ of $G - C$ which σ' maps to $S' \setminus D'$ and for which G contains a J - C edge. Let $\mathcal{J}$ be the set of all those components J . We shall group them by the way they attach to C , and define σ'' for these groups in turn.

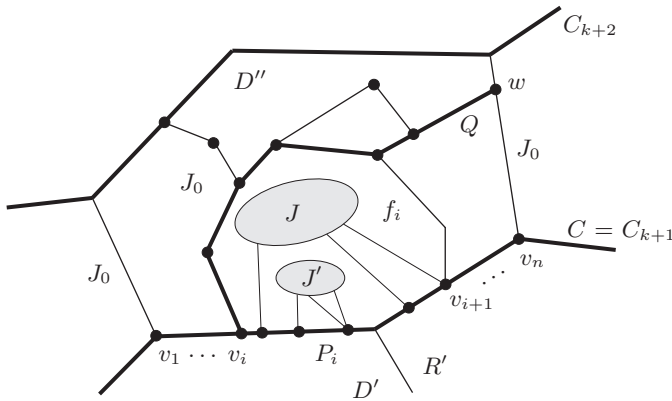


Fig. 12.7.5. Define σ'' jointly for the components $J, J' \in \mathcal{J}$ that attach to the same $P_i \subseteq C$

D'' P_i f_i

Since $m \geq k+2$, the disc D' lies inside a larger disc in S' , which is the union of D' and closed discs D'' bounded by the images under $\sigma' \circ f$ of the hexagons in R_{k+1} . By definition of $\mathcal{J}$, the embedding σ' maps every $J \in \mathcal{J}$ to such a disc D'' (Fig. 12.7.5). On the path P in C such that $\sigma'(P) = \sigma'(C) \cap D''$ (which is the image under f of one or two consecutive edges on S_{k+1}), let $v_1, \dots, v_n$ be the vertices with a neighbour in J_0 , in their natural order along P , and write P_i for the segment of P from v_i to v_{i+1} . For any v_i with $1 < i < n$, pick a v_i - J_0 edge and extend it through J_0 to a path Q from v_i to C_{k+2} (which exists by definition of J_0); let w be its first vertex that σ' maps to the boundary circle of D'' . By Lemma 4.1.2 applied to $\sigma'(v_i Q w)$ and the two arcs joining $\sigma'(v_i)$ to $\sigma'(w)$ along the boundary circle of D'' , there is no arc through D'' that links $\sigma'(P_{i-1})$ to $\sigma'(P_i)$ but avoids $\sigma'(v_i Q w)$. Hence, every $J \in \mathcal{J}$ with $\sigma'(J) \subseteq D''$ has all its neighbours on C in the same P_i , and σ' maps J to the face f_i of the plane graph $\sigma'(G[J_0 \cup C]) \cap D''$ whose boundary contains P_i . We shall define σ'' jointly on all those $J \in \mathcal{J}$ which σ' maps to this f_i , for $i = 1, \dots, n-1$ in turn.

To do so, we choose an open disc D_i in $S \setminus R$ that has a boundary circle containing $\sigma(P_i)$ and avoids the image of σ'' as defined until now. Such D_i exists in a strip neighbourhood of $\sigma(C)$ in S , because components $J' \in \mathcal{J}$ attaching to a segment $P_j \neq P_i$ of C send no edge to P_i . Choose a homeomorphism φ_i from the boundary circle of f_i to that of D_i so that $\sigma|_{P_i} = (\varphi_i \circ \sigma')|_{P_i}$, and extend this to a homeomorphism φ_i from the closure of f_i in S' to the closure of D_i in S . For every $J \in \mathcal{J}$ with $\sigma'(J) \subseteq f_i$, and for all J - C edges of G , define σ'' as $\varphi_i \circ \sigma'$. $\square$

(1.7.3)
(12.4.2)
(12.6.3)

Proof of Corollary 12.7.3. By their minimality, the graphs in $\mathcal{K}_{\mathcal{P}(S)}$ are incomparable under the minor-relation. If their tree-width is bounded, then $\mathcal{K}_{\mathcal{P}(S)}$ is well-quasi-ordered by the minor relation (Theorem 12.4.2), and hence must be finite. So assume their tree-width is unbounded, and let r be as in Lemma 12.7.4. By Theorem 12.6.3, some $H \in \mathcal{K}_{\mathcal{P}(S)}$ has a grid minor large enough to contain H^r . By Proposition 1.7.3, H^r is a topological minor of H , contrary to the choice of r . $\square$

We finally come to the proof of the graph minor theorem itself. The complete proof would still fill a book or two, but we are well equipped now to get a good understanding of its main ideas and overall structure. For background on surfaces, we once more refer to Appendix B.

(12.1.3)
(12.2.1)
(12.4.2)
(12.6.3)

Proof of the graph minor theorem (sketch). We have to show that every infinite sequence

$$G_0, G_1, G_2, \dots$$

of finite graphs contains a good pair: two graphs $G_i \preceq G_j$ with $i < j$. We may assume that $G_0 \not\preceq G_i$ for all $i \geq 1$, since G_0 forms a good pair

with any graph G_i of which it is a minor. Thus all the graphs $G_1, G_2, \dots$ lie in $\text{Forb}_{\preccurlyeq}(G_0)$, and we may use the structure common to these graphs in our search for a good pair.

We have already seen how this works when G_0 is planar: then the graphs in $\text{Forb}_{\preccurlyeq}(G_0)$ have bounded tree-width (Corollary 12.6.4) and are therefore well-quasi-ordered by Theorem 12.4.2. In general, we need only consider the cases of $G_0 = K^n$: since $G_0 \preccurlyeq K^n$ for $n := |G_0|$, we may assume that $K^n \not\preccurlyeq G_i$ for all $i \geq 1$.

The proof now follows the same lines as above: again the graphs in $\text{Forb}_{\preccurlyeq}(K^n)$ can be characterized by their tree-decompositions, and again their tree structure helps, as in Kruskal's theorem, with the proof that they are well-quasi-ordered. But as in Wagner's theorem (7.3.4) for $n = 5$, the parts in these tree-decompositions are no longer constrained in terms of order now but in more subtle structural terms. Roughly speaking, for every n there exists a finite set $\mathcal{S}$ of surfaces such that every graph without a K^n minor has a tree-decomposition into parts each 'nearly' embeddable in one of the surfaces $S \in \mathcal{S}$; see Theorem 12.6.6. By a generalization of Theorem 12.4.2 – and hence of Kruskal's theorem – it now suffices, essentially, to prove that the set of all the parts in these tree-decompositions is well-quasi-ordered: then the graphs decomposing into these parts are well-quasi-ordered, too. Since $\mathcal{S}$ is finite, every infinite sequence of such parts has an infinite subsequence whose members are all (nearly) embeddable in the same surface $S \in \mathcal{S}$. Thus all we have to show is that, given any surface S , all the graphs embeddable in S are well-quasi-ordered by the minor relation.

This is shown by induction on the Euler genus of S , using the same approach as before: if $H_0, H_1, H_2, \dots$ is an infinite sequence of graphs embeddable in S , we may assume that none of the graphs $H_1, H_2, \dots$ contains H_0 as a minor. If $S = S^2$ we are back in the case that H_0 is planar, so the induction starts. For the induction step we now assume that $S \neq S^2$. Again, the exclusion of H_0 as a minor constrains the structure of the graphs $H_1, H_2, \dots$, this time topologically: each H_i with $i \geq 1$ has an embedding in S which meets some circle $C_i \subseteq S$ that does not bound a disc in S in no more than a bounded number of vertices (and no edges), say in $X_i \subseteq V(H_i)$. (The bound on $|X_i|$ depends on H_0 , but not on H_i .) Cutting along C_i and capping the hole(s), we obtain one or two new surfaces of smaller Euler genus. If the cut produces only one new surface S_i , then our embedding of $H_i - X_i$ still counts as a near-embedding of H_i in S_i (since X_i is small). If this happens for infinitely many i , then infinitely many of the surfaces S_i are also the same, and the induction hypothesis gives us a good pair among the corresponding graphs H_i . On the other hand, if we get two surfaces S'_i and S''_i for infinitely many i (without loss of generality the same two surfaces), then H_i decomposes accordingly into subgraphs H'_i and H''_i embedded in these surfaces, with $V(H'_i \cap H''_i) = X_i$. The set of all these subgraphs taken

together is again well-quasi-ordered by the induction hypothesis, and hence so are the pairs (H'_i, H''_i) by Lemma 12.1.3. Using a sharpening of the lemma that takes into account not only the graphs H'_i and H''_i themselves but also how X_i lies inside them, we finally obtain indices i, j not only with $H'_i \preceq H'_j$ and $H''_i \preceq H''_j$, but also such that these minor embeddings extend to the desired minor embedding of H_i in H_j – completing the proof of the graph minor theorem.

The graph minor theorem does not extend to graphs of arbitrary cardinality, but it might extend to countable graphs. Whether or not it does appears to be a difficult problem. It may be related to the following intriguing conjecture, which easily implies the graph minor theorem for finite graphs (Exercise 68). Call a graph H a *proper minor* of G if there is a contraction from a subgraph of G onto H that is not an isomorphism from G to H .

Self-minor conjecture. (Seymour 1980s)

Every countably infinite graph is a proper minor of itself.

In addition to its impact on ‘pure’ graph theory, the graph minor theorem has had far-reaching algorithmic consequences. Using their structure theorem for the graphs in $\text{Forb}_{\preceq}(K^n)$, Theorem 12.6.6, Robertson and Seymour have shown that testing for any fixed minor is ‘fast’: for every graph H there is a polynomial-time algorithm¹⁴ that decides whether or not the input graph contains H as a minor. By the minor theorem, then, every minor-closed graph property $\mathcal{P}$ can be decided in polynomial (even cubic) time: if $\mathcal{K}_{\mathcal{P}} = \{H_1, \dots, H_k\}$ is the corresponding set of forbidden minors, then testing a graph G for membership in $\mathcal{P}$ reduces to testing the k assertions $H_i \preceq G$.

The following example gives an indication of how deeply this algorithmic corollary affects the complexity theory of graph algorithms. Let us call a graph *knotless* if it can be embedded in $\mathbb{R}^3$ so that none of its cycles forms a non-trivial knot. Before the graph minor theorem, it was an open problem whether knotlessness is decidable, that is, whether *any* algorithm exists (no matter how slow) that decides for any given graph whether or not that graph is knotless. To this day, no such algorithm is known. The property of knotlessness, however, is easily ‘seen’ to be closed under taking minors: contracting an edge of a graph embedded in 3-space will not create a knot where none had been before. Hence, by the minor theorem, there *exists* an algorithm that decides knotlessness – even in polynomial (cubic) time!

However spectacular such unexpected solutions to long-standing problems may be, viewing the graph minor theorem merely in terms of its corollaries will not do it justice. At least as important are the

¹⁴ indeed a cubic one – although with an enormous constant depending on H